

EECS3101: DESIGN AND ANALYSIS OF ALGORITHMS

Section Z – Winter 2026

SUBJECT TO CHANGES UNTIL: MONDAY, JANUARY 19

LAST UPDATED: JANUARY 5

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1 COURSE POLICIES

1. Assignments

You will receive **full** marks as long as successful submissions are made by the corresponding **submission deadlines**, which are enforced **strictly**.

The format of your programming assignments is **identical** to that of the subsequent programming tests. Therefore, it is in your best interest to follow the **export and submission process** (for which you will be expected to complete alone during the programming tests) and submit work that:

- complies with the instructions (so as to avoid penalties when you submit for grading in the actual programming tests); and
- represents your **true** and **best** attempt.

The rationales of this policy are that: 1) you can **rest assured that you will not lose any marks from assignments (as long as you submit them by the deadlines)**; and 2) **you can just focus on the learning by seeking help from colleagues, TAs, and your section instructor without worrying about violating the academic honesty policy**.

Please do not abuse this policy: **you are still 100% responsible for acquiring the intended understandings and skills from these assignments**. Be advised that **later scheduled (written and programming) tests, as well as the final written exam, will be based on these assignments**, so if you chose **not** to learn the materials responsibly (e.g., relying much on your colleagues, submitting incomplete work and only intending to look at solutions when they are made available), you would risk **poor performance** in subsequent tests and the exam.

2. ASSIGNMENT/TEST DEADLINES

Stringent deadlines are imposed on all scheduled in-person **written tests**, scheduled in-person **programming tests**, and **assignments** (to be submitted remotely via the *web submit* to the EECS server).

An in-person, written **exam** will be scheduled by the registrar office to take place during the **exam period**.

It is your **sole** responsibility to meet all these deadlines.

3. Test Dates and Locations

For each test, it is your **sole** responsibility for ensuring that you are available to take the test (during the chosen lecture or tutorial time of your **enrolled section**).

The exact time and location of each test will be announced at least one week prior to each test.

4. No Team Work Allowed for Scheduled Tests

All **written & programming tests** are to be completed **individually**.

When a scheduled test takes place between different sections, until all test sessions conclude, it is considered **a violation of academic integrity** if you communicate in any way, shape, or form with others about the test(s) already given.

5. **Plagiarism**: When submitting each of your **written tests** and **programming tests**, you claim that it is **solely** your work. It is considered as **a violation of academic integrity** if you copy or share **any** parts of your work (e.g., code, notes) during **any** stage of your development. The instructor and TAs may examine all submissions, and suspicious ones will be reported *immediately* to Lassonde as *a breach of academic integrity*. **We do not tolerate academic dishonesty.**
6. **Missed Tests**
- It is your **sole** responsibility for initiating the communication by the set deadline, or a grade penalty (e.g., a zero for the test) may be applied.
- If you missed a (programming or written) test:
- Contact Jackie by the end of the test day in order to gain the approval for accommodation.
 - Approval is at Jackie's discretion and should **not** be assumed to be automatic.
 - Once approved, if a **makeup test** is scheduled, it will be the only makeup opportunity that can be offered: you are supposed to prioritize attending it.
7. **Official Accommodation**
- Contact Jackie by the end of the first week (**Friday, Jan 9**).
 - For each programming or written test, you will write them in an EECS lab machine, set up with the appropriate accommodation. Accordingly:
 - cancel bookings of all tests with the alternate exam centre;
 - keep your booking of the final exam; and
 - discuss with Jackie to confirm the arrangement.
8. **LATE ENROLMENT**: Students who are not yet officially registered should assume an eventual successful enrolment into the course and are responsible for: **1)** contacting the section instructor **within Week 1** for course information (e.g., lecture materials, assignments access and deadlines); and **2)** attending lectures, submitting assignments, and taking scheduled tests in time.

No assignment deadline extensions or deferred tests due to late enrolment will be accommodated.

2 INSTRUCTOR

- Chen-Wei (JACKIE) Wang
 - Contact: jackie@eecs.yorku.ca (<https://www.eecs.yorku.ca/~jackie/>)

Jackie believes that **in-person** communication is the *most effective* for attending to your questions/concerns related course materials and grading. When you receive slow or no responses to your email inquiries, it is most likely an indication that Jackie is happy to help you during his **in-person** office hours and/or appointments.

- Office Hours:
 - * 16:30 – 17:30, Mondays, Wednesdays, Thursdays (**in-person**)
 - * by appointments (Zoom or In-Person)
- Campus Office: Lassonde Building, Room 2043 [19, D5 in the Keele campus]
Virtual Office: <https://yorku.zoom.us/my/jackie.loves.oxford>

3 VENUES

- Lectures
 - 14:30 – 15:50, Mondays & Wednesdays
LAS B (Lassonde Building) [D5/19 on the Keele Campus]
- Tutorials
 - 14:30 – 15:50, Thursdays
 - SLH F (Stedman Lecture Halls) [D5/22 on the Keele Campus]
- Both lectures and tutorials will be conducted in person by Jackie.
- In circumstances beyond our control (e.g., extreme weather), an announcement will be made in advance if the session needs to move online (e.g., via Zoom or pre-recorded videos).

4 PREREQUISITES

- **General Prerequisites:** A cumulative grade point average (GPA) of 4.50 or better over all previously completed Major EECS courses. The GPA computation excludes all EECS courses that have a second digit 5, or are Co-Op/PEP courses.
- LE/EECS 2101 3.00 or LE/EECS 2011 3.00; SC/MATH 1090 3.00; SC/MATH 1310 3.00.
Course credit exclusion: LE/SC CSE 3101 3.00; LE/DIGT 2102 8.00.

5 ECLASS SITE

<https://eclass.yorku.ca/course/view.php?id=141527>

6 STUDY MATERIALS

- There will be no textbooks for this course. Study your instructor’s lecture materials:
 - The lectures page:
https://www.eecs.yorku.ca/~jackie/teaching/lectures/index.html#EECS3101_W26
- If you wish to review OOP in Java, consider the study materials for:
 - EECS2030 (Introductory OOP):
https://www.eecs.yorku.ca/~jackie/teaching/lectures/index.html#EECS1022_W21
 - EECS2030 (Advanced OOP):
https://www.eecs.yorku.ca/~jackie/teaching/lectures/index.html#EECS2030_F21
 - EECS2011 (Data Structures):
https://www.eecs.yorku.ca/~jackie/teaching/lectures/index.html#EECS2011_W22
- Here are some optional reference textbooks:
 - Data Structures and Algorithms in Java, 6th Edition (2014), Wiley
Michael T. Goodrich, Roberto Tamassia, Michael H. Goldwasser
 - Introduction to Algorithms, 4th Edition (2022), The MIT Press
Thomas H. Cormen, Charles E. Leiserson, Ronald L. Rivest, Clifford Stein
 - Algorithms, 4th Edition (2011), Addison-Wesley Professional
Robert Sedgewick, Kevin Wayne [<https://algs4.cs.princeton.edu/home/>]
 - Algorithm Design (2005), Pearson
Jon Kleinberg, Eva Tardos

7 COURSE DESCRIPTION

This course is intended to teach students the fundamental techniques in the design of algorithms and the analysis of their computational complexity. Each of these techniques is applied to a number of widely used and practical problems.

At the end of this course, a student will be able to:

- choose algorithms appropriate for many common computational problems;
- exploit constraints and structure to design efficient algorithms; and
- select appropriate tradeoffs for speed and space.

Weekly three-hour lectures and 1.5-hour scheduled mandatory tutorials.

Topics covered may include:

- a review of fundamental data structures,
- asymptotic notation,
- solving recurrences,
- sorting and order statistics,
- divide-and-conquer approaches,
- dynamic programming,
- greedy method,
- divide-and-conquer algorithms,
- amortization approaches,
- graph algorithms, and
- the theory of NP-completeness.

8 COURSE LEARNING OUTCOMES (CLOs)

Upon completion of the course, students are expected to be able to:

CLO1 Choose an appropriate algorithm to solve a given computational problem, and justify that choice.

CLO2 Design new algorithms using a variety of techniques (recursion, greedy algorithm, dynamic programming, backtracking).

CLO3 Prove correctness of an algorithm using pre- and post-conditions and loop invariants.

CLO4 Prove bounds on the running time of an algorithm.

CLO5 Apply standard graph algorithms to a variety of problems.

9 GRADING SCHEME

		SUBTOTAL
3 Assignments (A1: 1.5%; A2: 1.5%; A3: 2%)	5%	35%
Programming Test 1	15%	
Programming Test 2	15%	
Written Test	15%	65%
Exam (Cumulative)	50%	

10 COURSE EXPECTATIONS & LEARNING PHILOSOPHY

This course is taught with a particular emphasis on foundational reasoning, individual mastery, and professional standards. Students are expected to read the following open letter **carefully and early**, which explains how learning, assessment, and grading are approached in this course:

[https://www.eecs.yorku.ca/~wangcw/teaching/lectures/2026/W/
Letter-from-Jackie-Winter-2026.pdf](https://www.eecs.yorku.ca/~wangcw/teaching/lectures/2026/W/Letter-from-Jackie-Winter-2026.pdf)

11 FINAL EXAM: CUMULATIVE & SUBSTANTIAL

- Your final exam will be *cumulative*: it will cover all study materials.
 - It will be an opportunity for you to continually *synthesize* topics that are connected.
- Therefore, your final exam will be the most substantial grading component.
 - It assesses how competently you can apply the learned concepts and skills.
 - The best preparation is to constantly review and reflect on the covered topics.

12 MAPPING RAW MARKS TO LETTER GRADES

- For each grading unit, you will receive a **raw mark score** (not necessarily out of 100).
- The weighted sum of all grading units will be mapped to its letter grade.
 - Refer to the University's common **Grades and Grading Schemes**.
 - e.g., Say there are only two grading units: Exam (60%) and Lab1 (40%).
Receiving 150 marks (out of 200) for Exam and 2 marks (out of 3) for Lab1 leads to a letter grade B (based on the weighted sum $\frac{150}{200} \times 60 + \frac{2}{3} \times 40 \approx 71.7$).

13 EXPECTED WEEKLY WORKLOAD

- Lassonde’s recommendation is 3 – 4.5 hours per credit: *9 – 13.5 hours* for a 3.00 course.
- “In-Class” Hours:
 - In-Class Lectures [3 hours]
 - Optional: Office Hours, Problem Solving Sessions
- “Out-of-Class” Hours:
 - Completing Assignments, Studying for Lectures/Tests [6 to 10.5 hours]
- Given that this is a *foundational course*, it is **not unreasonable** that you find yourself needing more time to digest the materials and build the skills. The harder you work in this course, the easier you may find subsequent years and job interviews.

14 ATTENDANCE OF CLASSES

- There are 23 upcoming in-person lectures in total (2 classes × 12 weeks – first class). If the course structure includes scheduled in-person tutorials, the same attendance and engagement expectations apply equally to those sessions.
- Attending in-person sessions (on time & with focus) is necessary for your learning.
- While attendance is your responsibility, Jackie strongly encourages you to attend these in-person sessions:
 - Attendance will be taken **randomly** (via iClicker).
 ⇒ Attendance will be checked approximately once every one or two classes.
*In a class where attendance is taken, **one or more checks** may be conducted: your attendance of that class will **not** count if you miss any of the checks.*
 For example, if you wait for the first check to occur and choose to leave right afterwards, your attendance will not count as you may miss the subsequent check(s). In other words, *your attendance for a class will count **only if all** checks are completed.*
 - Each attendance check will be conducted briefly (e.g., for a few minutes) at some time between **5 minutes** after class starts and **5 minutes** before class ends.
 - **No** makeup attendance will be considered if you missed a check because you, e.g.,
 - * arrived late
 - * left early
 - * did not pay attention or was absent when the attendance check took place
 - Please **always** have the iClicker launched on your computer (Chrome or Safari web browsers recommended) or mobile device:
 - * To accommodate the **(extremely) rare** occurrences of a failed check-in, you must approach Jackie **at the end of the same class**.
 You will be accommodated for **a maximum of 2 classes**.
 ⇒ *You are **solely** responsible for resolving any technical issues that caused you to fail checking in via the installed iClicker.*

e.g., see: <https://mhe.my.site.com/iclicker/s/article/How-to-Troubleshoot-Your-Connection-to-the-iClicker-Student-App>

- Attendance is **not** part of the grading scheme.
 - **No** fixed bonus percentage is awarded.
 - Your attendance record will be used as a **reference point**.
 - At the end of the semester, if your raw mark (out of 100) is not close (i.e., > 0.5 mark away) to the cutoff of the next letter grade, Jackie **may** exercise discretion in determining your final grade, considering factors such as:
 - * performance on the final exam;
 - * attendance record; and
 - * other evidence of consistent effort and engagement.
 - This discretionary consideration:
 - * may **not** guarantee any adjustment to the final letter grade.
 - * may **not** consider individual excuses (including illness or family emergency).
 - This is because attendance is **not** a graded component of the course.
 - Rather, it serves only as a coarse indicator of **presence**.
 - Because attendance is **not** part of the formal grading scheme, a missed class for any reason does **not** have any negative impact upon your grade.
 - Instead, attendance is used only as **one** of several possible reference points in borderline cases where discretion is applied (e.g., when a student’s raw mark is close to, but not quite at, the cutoff for the next letter grade).
 - In other words, missing classes will **not** “cost” you marks, but consistent attendance may strengthen the case for a positive discretionary decision at the end of the semester.
- Therefore:**

- once the attendance-taking window expires, **no** late responses will be accepted;
- **no** reasons will be considered for missing attendance checks.

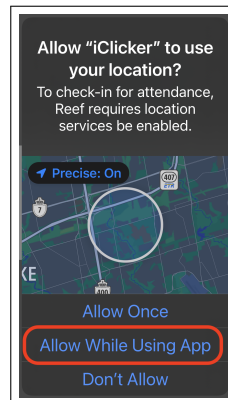
- Jackie reserves the right to **invalidate** your attendance records if, e.g.:
 - you wait only for attendance to be taken and leave the class shortly after;
 - you attend classes but cause distractions (e.g., talking, using devices for irrelevant activities) to Jackie and/or to other students.
- What should I do to set up the iClicker for attendance checks?
- Refer to this starter guide (to install iClicker on your mobile device):

<https://lthelp.yorku.ca/polling-students/iclicker-student-app-quick-start-guide>

When creating an iClicker account, be sure to supply your **student number** and **...@my.yorku.ca** email (you are responsible for **not** being considered for attendance if the student number or email is misplaced or misspelt).

- Ignore the first section “**For Courses using eClass integration**”.
- Follow these sections:
 - * “**For Courses not using eClass integration**”
 - * “**Add Your Instructor iClicker Course**”:
 - Search “EECS3101-Z (W’26) - Algorithms”*
 - * “**Respond to Polls**”

- When launching iClicker, it is critical that you allow iClicker to use your location; otherwise you will not be able to join the course and take attendance.



15 IN-PERSON SESSION PROTOCOL

While lectures are recorded for review and attendance is encouraged, **if you do decide to attend an in-person session (e.g., lecture, tutorial), you are expected to follow the rules below** to keep the classroom a respectful and focused space for learning.

- **Avoid side conversations during lecture** — regardless of how quietly you believe you are speaking, side conversations disrupt the instructor’s focus and the class.
- **Mobile phones are to be put away at all times**, except during the iClicker attendance check before the short break.
- **Headphones are to be put away at all times.**
- **Laptops are to be used only for taking notes or following along with lecture materials.** Using them to complete labs or assignments — whether for this course or for another — is disrespectful to both your instructor and fellow classmates.

When these rules are not followed, it distracts your instructor and your classmates.

- Your instructor does not have the energy to police the class every minute. When they intervene, it is because the behaviour has become disruptive enough for them to act.
- When you are approached, do not argue that “*others were doing the same thing*” or “*I was not doing other things on my laptop*” — **your instructor will address according to how they genuinely perceive the disruption in the moment, and that judgment is final.**
- **This section of the syllabus counts as a general warning.** In some cases, your instructor may give an individual reminder, but they may also directly ask a student to **leave without a further warning.** That decision is **not open to debate during class time.**
- Refusing to leave only wastes the time of your fellow students who come to focus and learn; if that happens, your instructor may involve **campus security** as a last resort to maintain a safe and focused learning environment.

These rules exist to protect the learning experience of students who attend class prepared and focused.

16 SEMESTER CALENDAR

Figure 1 summarizes the schedule of required work items:

- Each (programming or written) test will take place during the scheduled tutorial time.
- Specific details for each (programming or written) test will be announced in advance.

EECS3101 Design and Analysis of Algorithms (Section Z, Winter 2026) - Semester Calendar												
	MON		TUE		WED		THU		FRI			
	January 5		6		7		8		9			
Week 1	Lecture 1				Lecture 2		Tutorial 1					
Week 2	Lecture 3				Lecture 4		Tutorial 2					
Week 3	Lecture 5				Lecture 6		Tutorial 3					
Week 4	Lecture 7				Lecture 8		Tutorial 4					
Week 5	February 2		3		4		5		6		Assignment 1	
	Lecture 9		Release: A1		Lecture 10		Tutorial 5					
Week 6	Lecture 11				Lecture 12		Tutorial 6					
Reading Week	16		17		18		19		20			
			Due: A1									
Week 7	Lecture 13		Release: A2		Lecture 14		ProgTest1				Assignment 2	
Week 8	Lecture 15				Lecture 16		Tutorial 7					
Week 9	Drop Deadline		Due: A2		Lecture 17		Release: A3		Lecture 18		Assignment 3	
	Lecture 19				Lecture 20		Tutorial 8					
Week 10	Lecture 19				Lecture 20		WrittenTest					
Week 11	Lecture 21		Due: A3		Lecture 22		ProgTest2					
Week 12	Lecture 23				Lecture 24		Tutorial 9					
Week 13	6		7		Exam Period (April 8 to April 24)							
			Study Day									

Figure 1: EECS3101-E W26 Semester Calendar – Expected Work Items

17 COVERAGE OF TESTS

Tentatively, referencing the semester calendar in Figure 1 (p11):

- Written Test covers Lectures 1 – 18 and Tutorials 1 – 8
- Programming Test 1 covers Assignment 1
- Programming Test 2 covers Assignment 2 and Assignment 3

18 WEEKLY SCHEDULE

In the time table below, each cell denotes a 30-minute interval.

	Monday	Tuesday	Wednesday	Thursday	Friday
14:00					
14:30	EECS3101 Z		EECS3101 Z	EECS3101 Z	
15:00	Lecture		Lecture	Tutorial	
15:30	LAS B		LAS B	SLH F	
16:00					
16:30	Office Hour		Office Hour		
17:00	(In-Person)				
17:30					

19 (TENTATIVE) LECTURE TOPICS

Whereas the pace will be adjusted according to the class dynamics, the following topics are planned to be covered:

- Generics in Java: Linked Lists and Trees
- Growth of Functions and Asymptotic Running-Time Analysis
- Dynamic Arrays and Amortized Running-Time Analysis
- Balanced Binary Search Trees and Tree Rotations
- Priority Queue ADT, Heap Definitions, and Heap Operations
- Divide-and-Conquer Algorithms and Recurrence Relations
- Graph ADT: Terminology and Traversal Algorithms
- Graph ADT: Shortest Paths, Topological Sorts, and Spanning Trees
- Graph ADT: Implementation Strategies
- Greedy Algorithms and Loop Invariants for Iterative Algorithms
- Algorithm Correctness Arguments, including Mathematical Induction
- Dynamic Programming Principles
- NP-Completeness and Reductions (time permitting)

The order, depth, pacing, and emphasis of these topics may change as the semester progresses, depending on class progress, questions, and instructional needs. This list is **not** a contract, and it should not be interpreted as a guarantee of coverage depth, exam emphasis, or exclusion of related material discussed in lectures, tutorials, or assignments.